Problem Solving in Geometry (I)
20.07.2023

Warm Up

1. Let ABC be a triangle, M and N be the midpoints of AB and AC. If $BN = CM$, prove that $AB = AC$.

2. Let ABC be a triangle. D and E are the points of AB and AC, such that BE and CD are the angle bisectors of the triangle. If $BE = CD$, prove that $AB = AC$.

3. (IMO2003-2) Let ABC be an acute-angled triangle with $AB < AC$. Let Ω be the circumcircle of ABC. Let S be the midpoint of the arc CB of Ω containing A. The perpendicular from A to BC meets BS at D and meets Ω again at $E \neq A$. The line through D parallel to BC meets line BE at L. Denote the circumcircle of triangle BDL by ω. Let ω meet Ω again at $P \neq B$.
   Prove that the line tangent to ω at P meets line BS on the internal angle bisector of $\angle BAC$.